

1 Centered Rogers–Dougall resonance

Fix $\kappa, b, c > 0$ with $d_0 = 1 + 2\kappa - b - c > 0$, and write $x_n = n + \kappa$. For complex u with $\operatorname{Re} u < d_0$, put

$$W_n(u) = x_n \frac{\Gamma(n+2\kappa)\Gamma(n+b)}{\Gamma(n+1)\Gamma(n+1+2\kappa-b)} \frac{\Gamma(n+c)\Gamma(n+d_0-u)}{\Gamma(n+1+2\kappa-c)\Gamma(n+b+c+u)}, \quad (1.1)$$

$$P(u) = \frac{\Gamma(b)\Gamma(c)\Gamma(d_0-u)}{\Gamma(d_0)\Gamma(b+u)\Gamma(c+u)}. \quad (1.2)$$

Denominator Gamma factors always mean entire reciprocal-Gamma factors. The classical Rogers–Dougall sum gives $\sum_{n \geq 0} W_n(u) = \Gamma(u)P(u)/2$ for $0 < \operatorname{Re} u < d_0$ [1]. Its hypergeometric excess is $2u$.

Set $r = (\kappa, b - \kappa, c - \kappa, 1 + \kappa - b - c - u)$ and define

$$\begin{aligned} \ell_j(u) &= -\frac{\sum_{i=1}^4 B_{2j+1}(r_i)}{j(2j+1)}, \quad j \geq 1, \\ A_0(u) &= 1, \quad A_j(u) = \frac{1}{j} \sum_{h=1}^j h \ell_h(u) A_{j-h}(u). \end{aligned} \quad (1.3)$$

The centered Gamma form is $W_n(u) = x_n \prod_{i=1}^4 \Gamma(x_n + r_i) / \Gamma(x_n + 1 - r_i)$. Bernoulli reflection and the shifted Stirling expansion imply

$$W_n(u) = x_n^{-1-2u} \left(\sum_{j=0}^{K-1} A_j(u) x_n^{-2j} + O(x_n^{-2K}) \right). \quad (1.4)$$

Only even inverse-power corrections occur. The remainder is locally uniform in the parameters; fixed spectral derivatives introduce only powers of $\log x_n$ [2].

Proposition 1.1 (An ordinary convergent subtraction). *For $K \geq 1$ and $-K < \operatorname{Re} u < d_0$,*

$$\begin{aligned} \mathcal{R}_K(u) &:= \sum_{n=0}^{\infty} \left[W_n(u) - \sum_{j=0}^{K-1} A_j(u) x_n^{-1-2u-2j} \right] \\ &= \frac{1}{2} \Gamma(u) P(u) - \sum_{j=0}^{K-1} A_j(u) \zeta(1+2u+2j, \kappa). \end{aligned} \quad (1.5)$$

The bracketed series and every fixed spectral derivative converge normally in that strip. Apparent poles on the right side are removable.

Proof. Equation (1.4) gives a summable compact-uniform majorant $O(n^{-1-\epsilon}(1 + \log^m n))$ for each fixed derivative. On $0 < \operatorname{Re} u < d_0$, sum the components separately by classical Dougall summation. The identity theorem then gives (1.5) on the full strip, with removability supplied by its holomorphic left side. $\square$

Proposition 1.2 (Residue polynomial and complete collapse). *Let $N \geq 0$ and*

$$\rho_N = \frac{(-1)^N}{N!} (d_0)_N (1-b)_N (1-c)_N. \quad (1.6)$$

Then $A_N(-N) = \rho_N$, and

$$\mathcal{R}_{N+1}(-N) = \frac{\rho_N}{2} [\psi(N+1) - \psi(d_0 + N) - \psi(b) - \psi(c) + 2\psi(\kappa)]. \quad (1.7)$$

All displayed digamma arguments are positive. In particular, the formula also applies when $\rho_N = 0$.

Proof. Compare residues in (1.5). The Gamma term has residue $\rho_N/2$ at $u = -N$, and the only polar counterterm is $j = N$, with residue $A_N(-N)/2$. This proves the polynomial identity $A_N(-N) = \rho_N$.

The essential coefficient law is

$$(\partial_u - D)A_N(u) = \sum_{h=1}^N \frac{B_{2h}(\kappa)}{h} A_{N-h}(u), \quad D = \partial_\kappa + \partial_b + \partial_c. \quad (1.8)$$

Indeed, $Dd_0 = 0$ and $E = \partial_u - D$ acts on the four r_i by $Er_1 = -1$ and $Er_2 = Er_3 = Er_4 = 0$. Hence $E\ell_h = B_{2h}(\kappa)/h$. Apply E to the formal generating function $\sum A_j y^j = \exp(\sum \ell_j y^j)$ and extract y^N .

For generic b, c , the constant term of $\Gamma(-N+t)P(-N+t)/2$ is

$$\frac{\rho_N}{2} [\psi(N+1) - \psi(d_0 + N) - \psi(b - N) - \psi(c - N)].$$

The $j = N$ counterterm has constant term $A'_N(-N)/2 - \rho_N \psi(\kappa)$, while the lower counterterms sum to $-\sum_{h=1}^N B_{2h}(\kappa) A_{N-h}(-N)/(2h)$. Equation (1.8) cancels all of the latter terms, leaving $-D\rho_N/2$ in place of the coefficient derivative. Finally,

$$D\rho_N = \rho_N \sum_{h=1}^N ((b-h)^{-1} + (c-h)^{-1}).$$

The digamma shift relation proves (1.7). Continuity of the normally convergent series and of the final positive-argument formula proves the exceptional cases as well. $\square$

The full spectral germ retains reciprocal-Gamma zeros. Define

$$Q_N(t) = \frac{(-1)^N}{N!} \Gamma(1+t) \prod_{h=1}^N (1-t/h)^{-1} P(-N+t). \quad (1.9)$$

It is analytic at zero, $Q_N(0) = \rho_N$, and $\Gamma(-N+t)P(-N+t)/2 = Q_N(t)/(2t)$. With $a_j(t) = A_j(-N+t)$ and $\zeta(1+v, \kappa) = v^{-1} + \sum_{m \geq 0} (-1)^m \gamma_m(\kappa) v^m / m!$,

$$\begin{aligned} \mathcal{R}_{N+1}(-N+t) &= \frac{Q_N(t) - a_N(t)}{2t} - a_N(t) \sum_{m \geq 0} \frac{(-2)^m \gamma_m(\kappa)}{m!} t^m \\ &\quad - \sum_{j=0}^{N-1} a_j(t) \zeta(1 - 2(N-j) + 2t, \kappa). \end{aligned} \quad (1.10)$$

This is a holomorphic identity and therefore proves every Taylor coefficient formula. Only negative odd Hurwitz spectral arguments occur outside the Stieltjes part. To evaluate at a zero of a reciprocal Gamma factor, use

$$\frac{1}{\Gamma(w+t)} = \frac{\prod_{h=0}^{L-1} (w+h+t)}{\Gamma(w+L+t)}, \quad w+L > 0,$$

before expanding. A vanished summand is not necessarily a vanished jet.

At the interlaced point $u_* = -N - 1/2$, ordinary substitution instead gives

$$\mathcal{R}_{N+1}(u_*) = \frac{1}{2}\Gamma(u_*)P(u_*) + \sum_{j=0}^N A_j(u_*) \frac{B_{2(N-j)+1}(\kappa)}{2(N-j)+1}. \quad (1.11)$$

Its spectral jets involve $0, -2, \dots, -2N$, with no Stieltjes pole. These conclusions follow from the classical Bernoulli special values of Hurwitz zeta [3] and (1.5).

For $|z| < 1$, replace each counterterm by $A_j(u)\Phi(z, 1 + 2u + 2j, \kappa)$ and $W_n(u)$ by $W_n(u)z^n$. The same bracketed convergence theorem extends all fixed spectral jets continuously to $z = 1$. If $\kappa = p/q$, $1 \leq p \leq q$, then

$$\Phi(z, s, p/q) = q^{s-1} z^{-p/q} \sum_{\ell=0}^{q-1} e^{-2\pi i p \ell / q} \text{Li}_s(e^{2\pi i \ell / q} z^{1/q}),$$

initially for $0 < z < 1$. This gives exact polylogarithmic subtraction identities. At the endpoint, take the limit of the complete bracket, not of separately divergent pieces.

The classical Dougall sum is the input, not a new claim. The centered coefficient law and the explicit resonance/jet calculus are the continuation presented here. The larger article supplies detailed harmonic and primitive specializations, antecedent discussion, and reproducible diagnostics.

References

- [1] NIST Digital Library of Mathematical Functions, §16.4, Eq. 16.4.9, Rogers–Dougall very-well-poised sum. <https://dlmf.nist.gov/16.4>.
- [2] NIST Digital Library of Mathematical Functions, §5.11, Asymptotic expansions of the Gamma function. <https://dlmf.nist.gov/5.11>.
- [3] NIST Digital Library of Mathematical Functions, §25.11, Hurwitz zeta function. <https://dlmf.nist.gov/25.11>.